

FILTERED INERTIA AND ARTIN CONDUCTORS OF THE 27 LINES ON AN EXPLICIT CUBIC SURFACE

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ABSTRACT. We determine the complete lower filtered inertia of the degree-27 line field of an explicit smooth cubic surface whose normal closure has Galois group $W(E_6)$. Factor degrees in the 27-line action do not distinguish the required subgroup embeddings, and the wild primes 3 and 5 require higher ramification data. We combine exact maximal-order decompositions with the simultaneous actions on the 27 lines and the 36 double-sixes, exhaust the compatible subgroup classes, solve the branch-different equations, and apply Serre's graded tame-action law only after the finite alternatives have been enumerated. At 3 the filtration has orders $(18, 9, 3, 3, 3, 3, 3, 1)$, while at 5 it has orders $(20, 5, 5, 5, 1)$; the corresponding Artin conductor pairs on the nontrivial 6- and 20-dimensional constituents are $(11, 35)$ and $(7, 29)$. Writing $A = 181 \cdot 997 \cdot 2346241$ and $B = 283 \cdot 1801 \cdot 14932047182473291995860108491583652133938007263719$, we obtain

$$\mathrm{Disc}(E) = 3^{46} 5^{36} A^{18} B^6, \quad \mathrm{Disc}(K) = 3^{106560} 5^{80352} A^{34560} B^{25920}.$$

Two decomposition groups of orders 18 and 36 remain possible at 3, but they have the same filtered inertia and hence the same stated invariants. No decomposition Frobenius, bad Euler factor, epsilon factor, or root number is determined.

1. INTRODUCTION

The 27 lines on a smooth cubic surface carry a rigid $W(E_6)$ configuration, but a global identification of the line-field Galois group does not determine how that group ramifies. A bad-prime list records where ramification may occur, and a field discriminant records a weighted total. Neither datum identifies the embedded inertia subgroup, its higher ramification groups, or the Artin conductors of the irreducible constituents of the line permutation representation.

This paper resolves that local problem for the explicit surface [equation \(2.1\)](#). Its degree-27 line field E has normal closure K with $\mathrm{Gal}(K/\mathbf{Q}) = W(E_6)$. We determine the complete lower filtration at every finite ramified prime, including the wild primes 3 and 5, and then compute the Artin and Swan conductors on

$$\mathbf{Q}[27] = \mathbf{1} \oplus V_6 \oplus V_{20}.$$

The calculation closes globally with exact factorizations of $\mathrm{Disc}(E)$ and $\mathrm{Disc}(K)$ and with the archimedean plus/minus multiplicities of (V_6, V_{20}) .

The main obstruction is subgroup ambiguity. At a rational prime, factor degrees of the line polynomial describe the orbits of a decomposition group on 27 points. Several nonconjugate subgroups of $W(E_6)$ can have the same orbit partition. Even after a second 36-point action on the double-sixes is introduced, the prime 3 leaves three raw subgroup hits. Higher differentials and the action of tame inertia on the last wild grade are needed to finish the classification. We therefore use the two actions asymmetrically: the 27-line action carries the representations of interest, while the 36-double-six action separates embeddings. Only one frozen degree-36 polynomial certifies the local double-six partitions; the other is explicitly a bounded nonresult and is never corroborating evidence.

The decisive 3-adic step is finite and exhaustive. The simultaneous action scan gives subgroup classes 140, 142, 206. Normality and cyclic residue quotients leave four ordered (D, I) pairs. Across those pairs, every possible deep C_3 has Table-of-Marks profile 6, 7,

or 8, with profile 6 occurring twice. Exact branch-different equations give formal layer solutions

$$(7, -18), \quad (1, 6), \quad (7, -18).$$

Only profile 7 is nonnegative. Serre’s graded conjugation formula at the last, odd grade then forces inversion, selecting inertia class 140 and excluding the central class 142. Two decomposition overgroups remain:

$$(D, I) = (140, 140) \quad \text{or} \quad (206, 140).$$

They have the same filtered inertia and therefore the same conductor and discriminant consequences.

The complete output is summarized in [table 2](#) and [figure 1](#). In particular, the wild lower-order sequences are

$$p = 3 : (18, 9, 3, 3, 3, 3, 3, 1), \quad p = 5 : (20, 5, 5, 5, 1),$$

and their Artin pairs on (V_6, V_{20}) are $(11, 35)$ and $(7, 29)$. If

$$q = 14932047182473291995860108491583652133938007263719,$$

$$A = 181 \cdot 997 \cdot 2346241,$$

$$B = 283 \cdot 1801 \cdot q.$$

then

$$\text{Disc}(E) = 3^{46} 5^{36} A^{18} B^6, \quad \text{Disc}(K) = 3^{106560} 5^{80352} A^{34560} B^{25920}.$$

Contributions. The paper makes four instance-specific claims.

- (1) We separate the nine-prime surface divided-discriminant envelope from the exact eight-prime ramified support of both E and K , and give every maximal-order local row that is used.
- (2) We exhaust all $p = 3$ and $p = 5$ decomposition/inertia candidates. At 3, the complete deep- C_3 rational calculation and Serre’s odd-grade action determine one filtered inertia while preserving the two possible decomposition overgroups.
- (3) We identify the tame C_3 and root-reflection classes, compute every local fixed space and Artin–Swan pair on (V_6, V_{20}) , and derive the exact representation conductors and both field discriminants.
- (4) We determine the real signature and distinguish the Table-of-Marks subgroup index from the character-table element index at infinity.

These claims are bounded to the displayed surface. We do not claim a first result for arbitrary cubic surfaces, and the finite-group and local-field tools themselves are classical.

1.1. Context and closest precedent. Serre’s treatment of ramification groups supplies three universal ingredients used here: Krasner–Hensel factor stability, the conjugation action on G_i/G_{i+1} , and the conductor–discriminant identity [8]. Our use of the graded action is narrow: Chapter IV, §2, Proposition 9 forces inversion at grade 7, but it supplies neither the subgroup scan nor any local data for the frozen surface. Similarly, the conductor formulas convert proved fixed spaces and filtrations into exponents; they do not determine those inputs.

For cubic surfaces, Elsenhans and Jahnel give the closest explicit precedent. Their experiments include full- $W(E_6)$ examples, ramification-support calculations, and p -adic factor patterns [4, Propositions 21–22, pp. 651–652]. The present paper does not rebrand the computation of ramified primes as new. It adds, for a different frozen surface, the complete higher inertia at both wild primes, the local Artin–Swan data of (V_6, V_{20}) , and the normal-closure discriminant. Their separate study of the cubic-surface discriminant provides determinant-character context [5, Theorem 2.12], while the explicit maximal-order and higher-group calculations here are independent instance data.

The geometric reflection step belongs to the Picard–Lefschetz framework of SGA 7 II [3]. Saito’s divided-discriminant and determinant formulas make the hypersurface specialization precise [7]. Neither source identifies our three singular fibers. We verify the unique singular point, Hessian unit, critical Hensel lift, critical-value congruence, and valuation-one smoothing separately at each prime before invoking the universal reflection theorem. Published $W(E_6)$ trace conventions, including the corrected finite-field table of Banwait, Fité, and Loughran [1], are used only to keep group labels aligned; every relevant orbit and fixed space is reconstructed from the labelled actions.

TABLE 1. Boundary with the closest explicit cubic-surface precedent. The table compares mathematical scope, not software implementations.

question	explicit precedent	this paper
bad-prime and p -adic factor information	present for other full- $W(E_6)$ examples	exact surface envelope, maximal-order rows, and field support for equation (2.1)
higher wild filtration	not supplied by the cited precedent	complete lower groups at 3 and 5
constituent conductors	not supplied by the cited precedent	all local Swan/Artin pairs on (V_6, V_{20})
field discriminants	cubic discriminant context	exact $\text{Disc}(E)$ and normal-closure $\text{Disc}(K)$

1.2. Scope and organization. The decomposition ambiguity at 3 is recorded rather than guessed. Accordingly, the paper proves no decomposition Frobenius, bad Euler polynomial or factor, epsilon factor, or root number; it also makes no holomorphy, automorphy, analytic-continuation, or functional-equation claim. This firewall appears formally in [remark 2.3](#) and again in [section 7](#).

[Section 2](#) fixes the surface, representations, and theorem. [Section 3](#) establishes the local arithmetic and the two carrier roles. [Sections 4](#) and [5](#) classify the wild and tame inertia. [Section 6](#) derives the conductors and discriminants, and [section 7](#) treats infinity and the exact nonclaims. The appendices contain full arithmetic summaries, group tables, reflection witnesses, the validation map, and a source-use ledger.

2. THE SURFACE, ITS LINE FIELD, AND THE MAIN THEOREM

2.1. The frozen cubic and its two fields. In homogeneous coordinates (u_0, u_1, u_2, u_3) , let $Y = V(F) \subset \mathbf{P}_{\mathbf{Q}}^3$, where

$$\begin{aligned}
 F = & 75081586157u_0^3 - 28576620789u_0^2u_1 - 122000922135u_0^2u_2 - 5364921951u_0^2u_3 \\
 & + 164150208636u_0u_1^2 - 415458334296u_0u_1u_2 + 151070718312u_0u_1u_3 \\
 & + 1158143874300u_0u_2^2 + 114691988016u_0u_2u_3 + 113572676646u_0u_3^2 \\
 & + 6898957820u_1^3 + 1132596902196u_1^2u_2 - 30413540316u_1^2u_3 \\
 & - 2054867641020u_1u_2^2 + 151980984216u_1u_2u_3 + 36794420832u_1u_3^2 \\
 & + 2646295985484u_2^3 + 560186573940u_2^2u_3 + 706181383584u_2u_3^2 + 1884468968u_3^3.
 \end{aligned}
 \tag{2.1}$$

The coefficients have content one and the leading coefficient is positive, so [equation \(2.1\)](#) fixes the scalar convention. Exact Jacobian elimination shows that Y is smooth. On the standard Grassmann chart U_{01} , a line is the row span of

$$\begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{pmatrix}.$$

Restricting F to this line gives four equations. Their quotient is a field of degree 27, presented by a primitive irreducible eliminant $g(d) \in \mathbf{Z}[d]$. We write

$$E = \mathbf{Q}[d]/(g), \quad K = \text{the normal closure of } E.$$

The full coefficient vector of g , the rational back-substitutions for a, b, c , and the oriented field identity used by the maximal-order calculation are bound in the exact input record described in [appendix A](#). Printing the several-page coefficient vector here would add no local argument; [equation \(2.1\)](#) and the exact chart construction specify the geometric object.

The line-incidence pairing embeds the Galois action into the Weyl group of the E_6 root lattice. An irreducibility certificate for g , a modular order-five class, and an exact Picard-lattice enumeration give the frozen input

$$[E : \mathbf{Q}] = 27, \quad \text{Gal}(K/\mathbf{Q}) = W(E_6), \quad |W(E_6)| = 51840. \quad (2.2)$$

This paper does not recompute the group from ramification; instead, it determines the ramification inside the already identified group. The rational permutation module of the lines decomposes as

$$\mathbf{Q}[27] = \mathbf{1} \oplus V_6 \oplus V_{20}. \quad (2.3)$$

The 27-point carrier is where the two nontrivial representations live. A second permutation carrier, the 36 double-sixes on $Y_{\overline{\mathbf{Q}}}$, separates subgroup classes that have the same line-orbit data.

2.2. Ramification notation. Fix a finite prime p , an embedding $K \hookrightarrow \overline{\mathbf{Q}}_p$, a decomposition group $D_p \leq W(E_6)$, and inertia $I_p \trianglelefteq D_p$. We use the lower numbering

$$I_0 = I_p \supseteq I_1 \supseteq I_2 \supseteq \cdots.$$

For a rational representation V of $W(E_6)$, put

$$\text{Sw}_p(V) = \sum_{i \geq 1} \frac{|I_i|}{|I_0|} \text{codim } V^{I_i}, \quad (2.4)$$

$$a_p(V) = \text{codim } V^{I_0} + \text{Sw}_p(V). \quad (2.5)$$

These formulas use inertia only. In contrast, a bad Euler polynomial would require decomposition data beyond the filtered groups determined below.

Set

$$\begin{aligned} q &= 14932047182473291995860108491583652133938007263719, \\ A &= 181 \cdot 997 \cdot 2346241 = 423395612137, \\ B &= 283 \cdot 1801 \cdot q. \end{aligned} \quad (2.6)$$

Table-of-Marks labels always refer to the 350 subgroup classes in the GAP table of marks for $\mathbf{U4}(2).2$. They are reproducible identifiers rather than intrinsic group names.

2.3. Main theorem.

Theorem 2.1 (Complete filtered inertia and global closure). *For the cubic surface [equation \(2.1\)](#), the degree-27 line field E , and its normal closure K , the following statements hold.*

- (a) *The nine-prime surface divided-discriminant envelope is*

$$\mathcal{B}_Y = \{2, 3, 5, 181, 283, 997, 1801, 2346241, q\},$$

whereas the finite ramified support of both E and K is exactly

$$\mathcal{R}_{E,K} = \{3, 5, 181, 283, 997, 1801, 2346241, q\}.$$

In the displayed order on $\mathcal{B}_Y$,

$$(v_p(\text{Disc } E))_p = (0, 46, 36, 18, 6, 18, 6, 18, 6).$$

- (b) *At $p = 3$, inertia is subgroup ToM 140, with*

$$I_0 \cong (C_3^2) : C_2, \quad I_1 = C_3^2, \quad I_2 = \cdots = I_7 = C_3, \quad I_8 = 1,$$

and the deep C_3 is ToM 7. The two and only two surviving ordered pairs are

$$(D, I) = (140, 140) \quad \text{or} \quad (206, 140).$$

Thus $|D_3| \in \{18, 36\}$ is unresolved, and ToM 206 is a decomposition overgroup only. At $p = 5$,

$$D = I_0 = \text{ToM } 147 \cong C_5 : C_4, \quad I_1 = I_2 = I_3 = C_5, \quad I_4 = 1.$$

At 181, 997, 2346241, inertia is tame ToM 6 $\cong C_3$. At 283, 1801, q , inertia is the tame root-reflection subgroup ToM 2 $\cong C_2$.

- (c) *The local Swan and Artin pairs are*

p	$\text{Sw}_p(V_6, V_{20})$	$a_p(V_6, V_{20})$
3	(5, 18)	(11, 35)
5	(3, 12)	(7, 29)
181, 997, 2346241	(0, 0)	(6, 12)
283, 1801, q	(0, 0)	(1, 5).

Consequently

$$N(V_6) = 3^{11}5^7A^6B, \quad N(V_{20}) = 3^{35}5^{29}A^{12}B^5.$$

- (d) *The field discriminants are*

$$\text{Disc } E = 3^{46}5^{36}A^{18}B^6, \quad \text{Disc } K = 3^{106}5^{60}A^{80}B^{352},$$

and $N(V_6)N(V_{20}) = \text{Disc } E$.

- (e) *The signature of E is $(r_1, r_2) = (3, 12)$. The subgroup generated by complex conjugation is the Table-of-Marks subgroup ToM 5; separately, complex conjugation belongs to character-table element-class index 17 in `CharacterTable("U4(2).2")` under `CTblLib` 1.3.1. The class has size 540 and centralizer order 96, and*

$$V_6 : (d^+, d^-) = (3, 3), \quad V_{20} : (d^+, d^-) = (11, 9).$$

The proof is local-to-global. [Propositions 3.1](#) and [3.2](#) supply exact local arithmetic and the two action carriers. [Propositions 4.1](#) to [4.3](#) prove the wild 3-adic classification, while [propositions 5.1](#) to [5.4](#) treat the remaining ramified primes. The character calculation and both discriminants appear in [section 6](#); infinity is handled in [section 7](#).

Remark 2.2 (Decomposition is not inertia). The notation (D, I) is essential. The alternatives are $(140, 140)$ and $(206, 140)$, and neither alternative makes ToM 206 an inertia group. Every invariant in [theorem 2.1](#) depends on the common filtered inertia. No argument below selects between the two decomposition groups.

TABLE 2. Complete local theorem for the frozen line field. The $p = 3$ entry has two possible decomposition overgroups but one filtered inertia. Reflection-prime input is an ordinary-double-point/Picard–Lefschetz certification; no (e, f) decomposition row is claimed.

p	local arithmetic or geometric input	lower orders $(I_i)_i$	inertia
2	surface envelope only; field unramified	(1)	1
3	$(3, 1, 3), (6, 1, 7), (9, 1, 18)^2$	$(18, 9, 3, 3, 3, 3, 3, 1)$	ToM 140
5	$(1, 1, 0)^2, (5, 1, 7)^3, (10, 1, 15)$	$(20, 5, 5, 5, 1)$	ToM 147
181, 997, 2346241	$(3, 1, 2), (3, 2, 2), (3, 6, 2)$ at each p	$(3, 1)$	ToM 6
283, 1801, q	ordinary double point/Picard–Lefschetz; no (e, f) row	$(2, 1)$	ToM 2

p	$\text{Sw}(V_6, V_{20})$	$a(V_6, V_{20})$	$(v_p \text{ Disc } E, v_p \text{ Disc } K)$
2	(0, 0)	(0, 0)	(0, 0)
3	(5, 18)	(11, 35)	(46, 106560)
5	(3, 12)	(7, 29)	(36, 80352)
181, 997, 2346241	(0, 0)	(6, 12)	(18, 34560)
283, 1801, q	(0, 0)	(1, 5)	(6, 25920)

At 3, $(D, I) = (140, 140)$ or $(206, 140)$; see section 4.

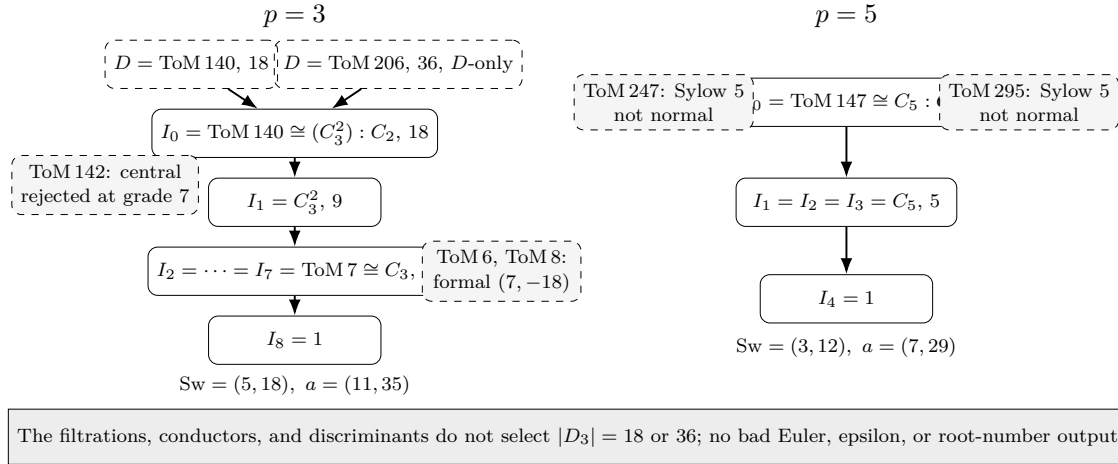


FIGURE 1. The two wild lower filtrations and the alternatives tested. At 3, branch-different equations select deep ToM 7; Serre’s odd-grade action then selects inertia ToM 140. At 5, normality and the three-layer equation select ToM 147.

Remark 2.3 (Exact scope firewall). NO-BAD-EULER-OR-ROOT-NUMBER: the theorem determines no decomposition Frobenius, bad Euler polynomial or factor, local epsilon factor, local or global root number, Artin holomorphy, automorphy, analytic continuation, or functional equation. Even a later resolution of D_3 would not by itself establish any of these independent assertions.

3. EXACT LOCAL ARITHMETIC AND THE TWO PERMUTATION CARRIERS

3.1. The surface envelope and field support. An exact Macaulay resultant calculation for the four partial derivatives of F , divided by the standard extraneous factor,

gives

$$\begin{aligned} \Delta_Y = & 2^{64} \cdot 3^{43} \cdot 5^7 \cdot 181^{24} \cdot 283 \cdot 997^{24} \cdot 1801 \\ & \cdot 2346241^{24} \cdot 14932047182473291995860108491583652133938007263719. \end{aligned} \quad (3.1)$$

The determinant was evaluated by two exact engines, one using fraction-free elimination and one using exact polynomial arithmetic. Their complete matrices and determinant guards are summarized in [appendix A](#). The factorization in [equation \(3.1\)](#) is a bad-reduction envelope for the integral surface model; it is not a field discriminant.

The global maximal order $\mathcal{O}_E$ has rank 27. Its exact discriminant has valuations

$$\begin{array}{c|cccccccccc} p & 2 & 3 & 5 & 181 & 283 & 997 & 1801 & 2346241 & q \\ \hline v_p(\text{Disc } E) & 0 & 46 & 36 & 18 & 6 & 18 & 6 & 18 & 6. \end{array} \quad (3.2)$$

Proposition 3.1 (Envelope and exact support). *The sets $\mathcal{B}_Y$ and $\mathcal{R}_{E,K}$ in [theorem 2.1](#) are respectively the nine-prime set supported by [equation \(3.1\)](#) and the eight-prime set obtained by deleting 2.*

Proof. Away from the support of the divided discriminant, the integral cubic has smooth reduction. The finite line scheme then extends as a finite étale scheme, so the Galois action on its geometric points is unramified. [Equation \(3.2\)](#) shows that all eight displayed odd primes ramify in E , while 2 does not.

It remains to pass from the non-Galois field E to its normal closure. At 2, the zero permutation conductor says that inertia fixes all 27 cosets of a line stabilizer $H \leq W(E_6)$. The kernel of this action is $\text{Core}_{W(E_6)}(H)$. Since the line action in [equation \(2.2\)](#) is faithful, this core is trivial. Thus inertia at 2 is trivial in K . The same faithfulness shows that nontrivial inertia at any of the eight odd primes cannot act trivially on all 27 lines. Hence E and K have the same stated support. $\square$

3.2. Local maximal-order rows. For a prime $\mathfrak{p} \mid p$ of E , write

$$e = e(\mathfrak{p}/p), \quad f = f(\mathfrak{p}/p), \quad d = v_{\mathfrak{p}}(\mathfrak{D}_{E_{\mathfrak{p}}/\mathbf{Q}_p}).$$

Its contribution to $v_p(\text{Disc } E)$ is fd . Exact prime-ideal decomposition in the maximal order gives the following complete rows.

TABLE 3. Maximal-order local rows. Every row in the last line occurs at each of the three rational primes, rather than being assigned one row per prime.

p	multiset of (e, f, d)	$\sum fd$	local degrees ef
3	$(3, 1, 3), (6, 1, 7), (9, 1, 18)^2$	46	$(3, 6, 9, 9)$
5	$(1, 1, 0)^2, (5, 1, 7)^3, (10, 1, 15)$	36	$(1, 1, 5, 5, 5, 10)$
181, 997, 2346241	$(3, 1, 2), (3, 2, 2), (3, 6, 2)$	18	$(3, 6, 18)$

The equalities

$$3 + 7 + 18 + 18 = 46, \quad 0 + 0 + 7 + 7 + 7 + 15 = 36, \quad 2 + 2 \cdot 2 + 6 \cdot 2 = 18$$

are direct different calculations, made before the representation-theoretic conductor computation and used later as checks. No constituent conductor is obtained by splitting a global sum.

At the reflection primes 283, 1801, q , we deliberately do not list an (e, f, d) decomposition. Their inertia is obtained by the geometric ordinary-double-point argument in [section 5.4](#). The six transpositions of a root reflection then give $v_p(\text{Disc } E) = 6$, consistently with [equation \(3.2\)](#).

3.3. The 36-point authority and its boundary. Let $\theta_{36} \in \mathbf{Z}[T]$ be the frozen degree-36 polynomial whose roots label the double-sixes. A second frozen polynomial δ_{36} has the same global splitting field but is not a local authority in this paper. The distinction matters because a stable-looking finite-precision factorization does not by itself identify the factors over $\mathbf{Q}_p$.

The factor-stability step uses the Krasner–Hensel criterion in the form localized in Serre [8, Chapter II, §2, Exercises 1–2, p. 30]. For each asserted factorization, the working precision exceeds both the valuation of the global polynomial discriminant and twice the largest factor-polynomial discriminant valuation; the monic factors are simple and multiply back to the original polynomial at the same precision. The source supplies the criterion, while all exponents, factors, and inequalities below are computations for [equation \(2.1\)](#).

Proposition 3.2 (Certified double-six partitions). *The polynomial θ_{36} is a certified local authority with the records in [table 4](#). No theorem assertion depends on δ_{36} , and no apparent δ_{36} factorization is used as corroboration.*

TABLE 4. Certified θ_{36} factor degrees and strict precision bounds. The two bounds in the fourth column are the global polynomial-discriminant valuation and twice the largest factor-polynomial discriminant valuation.

p	certified precisions	factor degrees	bounds	conclusion
3	(900, 950, 1000)	(3, 3, 3, 9, 18)	(886, 538)	all three precisions clear both
5	(900, 950, 1000)	(1, 5, 10, 10, 10)	(746, 246)	all three precisions clear both
181, 997, 2346241	(20, 30, 40)	(3, 6, 9, 18)	(24, 24)	precision 40 clears both

For comparison only, at the tame C_3 primes the global polynomial-discriminant valuation of δ_{36} is 840, and twice its largest factor-polynomial discriminant valuation is 408. Precision 40 is below both values. We therefore assign δ_{36} the formal role *bounded non-result and nondependency*. It supplies neither a premise nor a cross-check, at the tame or wild primes. Polynomial discriminants appear nowhere else in the proof: they are separation bounds, not substitutes for the field discriminants of E or K .

3.4. Why two actions are needed. For $D \leq W(E_6)$, its orbits on the 27 lines have sizes equal to the local factor degrees of g . Its orbits on the 36 double-sixes similarly match the certified factor degrees of θ_{36} . Thus the wild targets are

$$\begin{array}{c|cc}
 & 27\text{-line carrier} & 36\text{-double-six carrier} \\
 \hline
 p = 3 & (3, 6, 9, 9) & (3, 3, 3, 9, 18) \\
 p = 5 & (1, 1, 5, 5, 5, 10) & (1, 5, 10, 10, 10).
 \end{array} \tag{3.3}$$

The line carrier alone leaves multiple subgroup embeddings. The double-six carrier removes most of this ambiguity, but at 3 three raw decomposition classes still have the same simultaneous orbit targets. Normality, tame-quotient structure, branchwise differentials, and the graded tame action are therefore genuine proof steps, not database decoration.

The tame order-three case illustrates the same asymmetry. Two C_3 classes have compatible line behavior, but on double-sixes they have types 3^{12} and $1^3 3^{11}$. The certified θ_{36} type 3^{12} selects ToM 6; see [proposition 5.3](#). The 27-line action then supplies the (V_6, V_{20}) fixed spaces used for conductors.

4. WILD INERTIA AT $p = 3$

The prime 3 is the only place where the filtered inertia is unique but the decomposition group is not. We retain every finite alternative until it is removed by an explicit local-field condition. This order is important: Serre's conjugation law distinguishes the last two inertia embeddings, but it does not produce the candidate list.

4.1. Complete decomposition and inertia inventory. The local degrees in the line field and the certified double-six degrees are

$$27 : (3, 6, 9, 9), \quad 36 : (3, 3, 3, 9, 18).$$

We enumerate all 350 conjugacy classes of subgroups in the table of marks for $\mathbf{U4}(2) . 2$, reconstruct their actions on both labelled carriers, and retain precisely those with the two displayed orbit partitions.

Proposition 4.1 (Raw $p = 3$ hits). *The simultaneous orbit targets have exactly three subgroup hits:*

ToM index	$ H $	$\text{IdGroup}(H)$	initial role
140	18	(18, 4)	D or I
142	18	(18, 3)	D or I
206	36	(36, 10)	D only.

The complete list compatible with inertia normality and a cyclic residue quotient is

$$(D, I, |D/I|) = (140, 140, 1), (142, 142, 1), (206, 140, 2), (206, 142, 2). \quad (4.1)$$

Proof. The simultaneous orbit scan is exhaustive over the 350 classes, so it gives the three raw hits without a subgroup-name guess. Their 3-Sylow subgroup has order 9. If ToM 206 were inertia, the quotient of order 4 by wild inertia would be V_4 , whereas I_0/I_1 must be cyclic of order prime to 3. Hence ToM 206 cannot be I_0 . It contains normal subgroups in each of the two order-18 classes, both with cyclic quotient of order 2. Testing normality and the residue quotient for every ordered containment gives exactly [equation \(4.1\)](#). $\square$

The abstract group structures clarify the remaining ambiguity:

$$\text{ToM } 140 \cong (C_3^2) : C_2, \quad \text{ToM } 142 \cong C_3 \times S_3.$$

Both contain the required wild group $P = C_3^2$; they differ in how the tame involution acts on the eventual deep subgroup. The order-36 class has no role as inertia anywhere below.

4.2. Exhausting the deep C_3 . The four branches of $E \otimes \mathbf{Q}_3$ have target different vector

$$\mathbf{d} = (3, 7, 18, 18). \quad (4.2)$$

The tame I_0 contribution is $\mathbf{b} = (2, 5, 8, 8)$. One lower layer equal to $P = C_3^2$ contributes $\mathbf{u} = (1, 2, 4, 4)$. Let r be the number of P layers and s the number of layers equal to a chosen deep $Q \cong C_3$. For every candidate Q , the branch equations are

$$\mathbf{b} + r\mathbf{u} + s\mathbf{w}_Q = \mathbf{d}, \quad r, s \in \mathbf{Z}_{\geq 0}. \quad (4.3)$$

The deep group is not chosen by abstract isomorphism type: three different C_3 embeddings occur in the valid pairs. Their complete profile multiset is ToM 6 twice, ToM 7 once, and ToM 8 once. The two occurrences of ToM 6 come from distinct ordered (D, I) containments; neither is suppressed before the different equation is solved.

Proposition 4.2 (Unique branch-different solution). *[Equation \(4.3\)](#) has a nonnegative integral solution only for $Q = \text{ToM } 7$, and that solution is $(r, s) = (1, 6)$. Consequently*

$$I_1 = C_3^2, \quad I_2 = \cdots = I_7 = \text{ToM } 7 \cong C_3, \quad I_8 = 1.$$

TABLE 5. Exact deep- C_3 exhaustion. Orbit notation is run-length notation, and every fraction is exact.

Q	multiplicity	line type	double-six type	$\dim(V_6^Q, V_{20}^Q)$	$\mathbf{w}_Q$	formal (r, s)
ToM 6	2	3^9	3^{12}	$(0, 8)$	$(\frac{1}{3}, \frac{2}{3}, 1, 1)$	$(7, -18)$
ToM 7	1	$1^9 3^6$	$1^6 3^{10}$	$(4, 10)$	$(0, 0, 1, 1)$	$(1, 6)$
ToM 8	1	3^9	$1^3 3^{11}$	$(2, 6)$	$(\frac{1}{3}, \frac{2}{3}, 1, 1)$	$(7, -18)$

Proof. For ToM 6 and ToM 8, the first and third components of [equation \(4.3\)](#) give

$$r + \frac{s}{3} = 1, \quad 4r + s = 10,$$

whose unique formal integral solution is $(7, -18)$. It cannot be a filtration length. For ToM 7, the first component gives $r = 1$, and the third gives $8 + 4 + s = 18$, hence $s = 6$. The second and fourth components agree. The candidate list in [table 5](#) is exhaustive, so no fourth profile remains. $\square$

The fractional entries in [table 5](#) are not rounded orbit heuristics. They are the exact normalized contributions $|Q|/|I_0|$ to the four branch permutation conductors. The negative solution therefore rejects an embedding, not merely a proposed break number.

4.3. The odd-grade tame action. After [proposition 4.2](#), the only remaining choice is how the tame quotient $C_2 = I_0/I_1$ acts on the selected $Q = G_7/G_8 \cong C_3$. Serre's graded action law states that, for $s \in G_0$ and $\tau \in G_i/G_{i+1}$,

$$\theta_i(s\tau s^{-1}) = \theta_0(s)^i \theta_i(\tau). \quad (4.4)$$

Here $\theta_0(s) = -1$ for the nontrivial tame involution. The precise source is Serre [8, Chapter IV, §2, Proposition 9, pp. 69–70]. It supplies [equation \(4.4\)](#); it does not identify any Table-of-Marks class for the present surface.

Proposition 4.3 (Serre discriminator). *The inertia subgroup is ToM 140. The central embedding ToM 142 is impossible, and the final ordered pairs are exactly*

$$(D, I) = (140, 140), \quad (206, 140). \quad (4.5)$$

Proof. The last nonzero grade is $i = 7$. Substitution in [equation \(4.4\)](#) gives

$$\theta_7(s\tau s^{-1}) = (-1)^7 \theta_7(\tau) = -\theta_7(\tau),$$

so the tame involution acts by inversion on $Q \cong C_3$. In ToM 140, the selected normal ToM 7 is inverted. In ToM 142, it is centralized. The latter contradicts the graded action law. Removing the two rows with inertia 142 from [equation \(4.1\)](#) leaves precisely [equation \(4.5\)](#). The selected deep subgroup is normal in each surviving decomposition group. $\square$

Serre's proposition acts only after the 350-class scan, the four valid pairs, and the three exact deep-profile equations have been exhausted; it does not by itself select class 140.

4.4. Character consequences and the unresolved overgroup. The fixed dimensions for the three nontrivial layers are

H	$I_0 = \text{ToM 140}$	$P = C_3^2$	$Q = \text{ToM 7}$
$\dim V_6^H$	0	0	4
$\dim V_{20}^H$	3	4	10.

(4.6)

Thus the tame codimensions are $(6, 17)$, the P codimensions are $(6, 16)$, and the Q codimensions are $(2, 10)$. Direct substitution into [equation \(2.4\)](#) yields

$$\text{Sw}_3(V_6, V_{20}) = \frac{9}{18}(6, 16) + 6\frac{3}{18}(2, 10) = (5, 18), \quad (4.7)$$

$$a_3(V_6, V_{20}) = (6, 17) + (5, 18) = (11, 35). \quad (4.8)$$

Their sum 46 equals the direct different exponent in [table 3](#); [section 6](#) gives the general conductor argument and branchwise checks.

The two possibilities in [equation \(4.5\)](#) have orders 18 and 36. Both contain the same filtered inertia and the same labelled fixed spaces, so [equations \(4.7\)](#) and [\(4.8\)](#) and both field discriminants are independent of this choice. A decomposition Frobenius or a bad Euler polynomial would not be independent of decomposition data. We make no such inference here, and resolving $|D_3|$ alone would still not provide epsilon factors or root numbers.

5. WILD $p = 5$ AND THE SIX TAME PRIMES

5.1. The $p = 5$ decomposition pair. At 5, the two carrier partitions are

$$27 : (1, 1, 5, 5, 5, 10), \quad 36 : (1, 5, 10, 10, 10).$$

The simultaneous scan again begins with every subgroup class rather than a presumed Frobenius group.

Proposition 5.1 (Unique $p = 5$ pair). *The raw simultaneous hits and their exact small-group identifiers are*

ToM index	$ H $	$\text{IdGroup}(H)$	Sylow-5 normal?
147	20	(20, 3)	yes
247	60	(60, 5)	no
295	120	(120, 34)	no.

The only valid ordered triple is

$$(D, I, |D/I|) = (147, 147, 1), \quad I_0 \cong C_5 : C_4.$$

Proof. The three rows are the complete 350-class simultaneous orbit scan. The first ramification group I_1 is the unique wild Sylow-5 subgroup of inertia and must be normal in I_0 . The Sylow-5 subgroups of ToM 247 and ToM 295 are not normal, so neither class can be inertia. Normality and the cyclic residue-quotient test leave ToM 147; the orbit sizes already force $D = I$ for this class. $\square$

5.2. The $p = 5$ break equation. The six local branches have target different vector

$$\mathbf{d}_5 = (0, 0, 7, 7, 7, 15).$$

The $I_0 = C_5 : C_4$ base contribution is $\mathbf{b}_5 = (0, 0, 4, 4, 4, 9)$. Each positive layer C_5 contributes $\mathbf{w}_5 = (0, 0, 1, 1, 1, 2)$. Hence the only unknown layer count s satisfies

$$\mathbf{b}_5 + s\mathbf{w}_5 = \mathbf{d}_5. \quad (5.1)$$

Proposition 5.2 (The $p = 5$ filtration). *[Equation \(5.1\)](#) has the unique nonnegative integral solution $s = 3$. Thus*

$$I_0 = C_5 : C_4, \quad I_1 = I_2 = I_3 = C_5, \quad I_4 = 1.$$

Proof. Any of the three degree-5 branches gives $4 + s = 7$, and the degree-10 branch gives $9 + 2s = 15$. Both equations give $s = 3$; the two unramified branches remain zero. Since [proposition 5.1](#) left only one inertia embedding, no competing filtration remains. $\square$

The fixed dimensions are

$$\begin{array}{c|cc}
 H & I_0 = C_5 : C_4 & C_5 \\
 \hline
 \dim V_6^H & 2 & 2 \\
 \dim V_{20}^H & 3 & 4.
 \end{array} \tag{5.2}$$

Therefore

$$\mathrm{Sw}_5(V_6, V_{20}) = 3 \frac{5}{20} (4, 16) = (3, 12), \tag{5.3}$$

$$a_5(V_6, V_{20}) = (4, 17) + (3, 12) = (7, 29). \tag{5.4}$$

The sum $7 + 29 = 36$ recovers the direct different.

5.3. Tame order-three primes. At each of 181, 997, 2346241, the line carrier has inertia type 3^9 , which by itself is shared by more than one C_3 embedding. The double-six local rows are decisive. For ToM6, they have degrees $(3, 6, 9, 18)$, all with $e = 3$, and total different exponent 24; for the degree-only competitor, three double-sixes are fixed and the corresponding exponent is 22. Proposition 3.2 certifies the first set of rows from θ_{36} , with no use of δ_{36} .

Proposition 5.3 (Tame C_3 class). *At $p = 181, 997, 2346241$, inertia is $\mathrm{ToM}6 \cong C_3$. Its line type is 3^9 , its double-six type is 3^{12} , and*

$$\dim(V_6^{C_3}, V_{20}^{C_3}) = (0, 8), \quad a_p(V_6, V_{20}) = (6, 12).$$

Proof. The line decomposition rows permit two C_3 classes. The certified θ_{36} rows have double-six type 3^{12} , selecting ToM6 over the class with type $1^3 3^{11}$. Since the place is tame, the conductor is the codimension of the invariant space. The labelled line matrices give fixed dimensions $(0, 8)$, hence codimensions $(6, 12)$. $\square$

5.4. Reflection primes. The primes $283, 1801, q$ are treated geometrically. For each p , reduce the integral surface equation modulo p and split projective space into the four disjoint pivot strata

$$u_0 = \cdots = u_{j-1} = 0, \quad u_j = 1 \quad (0 \leq j \leq 3).$$

On each stratum, eliminate the four specialized homogeneous partial derivatives in the remaining coordinates. The calculation produces one reduced point on pivot stratum 0 and a unit ideal on strata 1, 2, 3. (Euler's identity, with $p \neq 3$, puts every common gradient zero on the surface.) At the unique point the three-by-three affine Hessian is invertible.

The invertible Hessian does two jobs. First, it proves that the special-fiber singularity is an ordinary quadratic singularity. Second, Hensel's lemma applied to the affine gradient gives a unique critical-point lift modulo p^2 . Direct substitution shows that the lifted critical value is congruent modulo p^2 to the value at the integer representative, and that $F(P)/p$ is nonzero modulo p . Thus the smoothing parameter has valuation exactly one. The total space is regular and meets the discriminant transversely.

For odd residue characteristic, the Picard–Lefschetz formula identifies the tame monodromy around such a transverse ordinary quadratic singularity with reflection in the vanishing root; see SGA 7 II, Exposé XV [3], and the cubic-surface determinant specialization in Saito [7, Proposition 2.3, p. 858; Theorem 3.5, pp. 864–866; cubic-surface specialization, p. 870]. These sources provide the universal implication. The singular points, Hessians, lifts, and valuation-one assertions are the exact instance calculations recorded in [appendix C](#).

The finite-group side is also exhaustive. There are four classes of subgroups generated by an involution, and their two carrier types are as follows.

TABLE 6. All order-two subgroup profiles. The element-index column belongs to the character table, not the table of marks.

ToM	line type	double-six type	$\dim(V_6^H, V_{20}^H)$	normalizer	element-class size	element index
2	$1^{15}2^6$	$1^{16}2^{10}$	(5, 15)	1440	36	16
3	1^32^{12}	$1^{12}2^{12}$	(2, 12)	1152	45	2
4	1^72^{10}	1^82^{14}	(4, 12)	192	270	3
5	1^32^{12}	1^42^{16}	(3, 11)	96	540	17

Proposition 5.4 (Reflection inertia). *At each of 283, 1801, q , inertia is the tame root-reflection subgroup ToM 2. Its Swan pair is $(0, 0)$, and its Artin pair is $(1, 5)$. This is geometric ordinary-double-point/Picard–Lefschetz certification; no (e, f) decomposition row is claimed.*

Proof. The exact chart, Hessian, Hensel, critical-value, and valuation computation proves the hypotheses of the odd-characteristic Picard–Lefschetz formula, so a tame generator acts as an E_6 root reflection. On the 27 lines, a root reflection has type $1^{15}2^6$; on the 36 double-sixes it has type $1^{16}2^{10}$. The exhaustive comparison in table 6 uniquely identifies ToM 2. The fixed dimensions (5, 15) give tame codimensions (1, 5), and tame inertia has zero Swan conductor. $\square$

It is tempting to reverse the argument and infer a local factor row from the six transpositions. We do not do so. The geometry determines the inertia conjugacy class and hence the permutation conductor 6; it does not determine the decomposition or residue degrees of the primes of E above a reflection prime.

6. ARTIN–SWAN CONDUCTORS AND THE TWO DISCRIMINANTS

6.1. Layerwise character calculation. The Artin formula equation (2.5) separates the tame codimension at I_0 from the positive layers. All fixed spaces are computed from the labelled 27-line action, not reconstructed from the total discriminant exponent. The data needed for every ramified type fit in one table.

TABLE 7. Fixed spaces and weights in the local conductor formula. A multiplicity indicates the number of lower layers with that subgroup.

type	subgroup H	multiplicity	weight $ H / I_0 $	$\dim V_6^H$	$\dim V_{20}^H$	codimensions
$p = 3$	$I_0 = \text{ToM } 140$	1	1	0	3	(6, 17)
	C_3^2	1	1/2	0	4	(6, 16)
	deep ToM 7 $\cong C_3$	6	1/6	4	10	(2, 10)
$p = 5$	$I_0 = \text{ToM } 147$	1	1	2	3	(4, 17)
	C_5	3	1/4	2	4	(4, 16)
tame C_3	ToM 6	1	1	0	8	(6, 12)
reflection	ToM 2	1	1	5	15	(1, 5)

At 3, the positive-layer contribution is

$$\frac{1}{2}(6, 16) + 6\frac{1}{6}(2, 10) = (5, 18),$$

and adding the I_0 codimensions gives (11, 35). At 5,

$$3\frac{1}{4}(4, 16) = (3, 12),$$

and adding (4, 17) gives (7, 29). The tame rows have no positive lower group, so their Swan conductors vanish and their Artin conductors are the final codimension column.

Proposition 6.1 (Local conductor table). *The local pairs are exactly those in [theorem 2.1\(c\)](#). In particular, every Swan conductor and every Artin conductor is a non-negative integer obtained layer by layer.*

Proof. The wild calculations are the exact weighted sums above. At each tame C_3 prime, [proposition 5.3](#) supplies fixed dimensions (0, 8), hence codimensions (6, 12). At each reflection prime, [proposition 5.4](#) supplies (5, 15), hence (1, 5). Outside the eight-prime support inertia is trivial. These cases exhaust all finite primes by [proposition 3.1](#). $\square$

6.2. Branchwise conductor–different closure. For a finite separable extension, the Artin conductor of its permutation representation is the discriminant exponent; see Serre [8, Chapter VI, §2, Corollary 1', pp. 100–101, and §3, Proposition 6 and Corollary 1, pp. 103–104]. Applied locally to each orbit of D_p on the 27 embeddings, this identifies the branch permutation conductor with $f_p d_p$.

Proposition 6.2 (Branchwise closure). *For every local branch at 3, 5, 181, 997, 2346241, the permutation conductor computed from the proved lower filtration equals the direct maximal-order different contribution. Summing the branches gives 46, 36, and 18 for the three prime types. At a reflection prime, the six transpositions give conductor 6, equal to the global maximal-order valuation.*

Proof. For the maximal-order primes, restrict the 27-point permutation action to each D_p orbit and apply the local conductor–different identity. The orbit refinements used in [equations \(4.3\)](#) and [\(5.1\)](#) give the direct rows in [table 3](#), rather than merely their totals. For a tame reflection the Artin conductor of a permutation is the codimension of its fixed subspace: six disjoint transpositions have 21 orbits on 27 points, so the codimension is $27 - 21 = 6$. [Equation \(3.2\)](#) supplies the independent global-order value. $\square$

The decomposition [equation \(2.3\)](#) now gives

$$a_p(V_6) + a_p(V_{20}) = v_p(\text{Disc } E) \quad (6.1)$$

at every p . We emphasize the direction of use: the fixed-space calculation proves the two summands, and [equation \(6.1\)](#) checks them. The equality is not used to guess one constituent from the other.

6.3. Global representation conductors and Disc E . Multiplying the local powers in [proposition 6.1](#) yields:

Corollary 6.3 (Global Artin conductors). *With A, B as in [equation \(2.6\)](#),*

$$N(V_6) = 3^{11}5^7 A^6 B, \quad N(V_{20}) = 3^{35}5^{29} A^{12} B^5.$$

Moreover,

$$N(V_6)N(V_{20}) = 3^{46}5^{36} A^{18} B^6 = \text{Disc } E. \quad (6.2)$$

Proof. At 3 and 5 use the exponents (11, 35) and (7, 29). Each of the three factors of A carries exponents (6, 12); each of the three factors of B carries exponents (1, 5). Addition of the two exponent vectors gives 46, 36, 18, 6 by prime type. The maximal-order discriminant has exactly these valuations and is positive, proving [equation \(6.2\)](#). $\square$

The exact field discriminant also closes the support argument. No finite prime outside the displayed eight can act nontrivially on the 27 roots. Because the $W(E_6)$ action on those roots is faithful, no hidden inertia can survive in K while disappearing from E .

6.4. The normal-closure discriminant. For a Galois extension with group G and lower inertia I_i , the exponent of the regular permutation discriminant is

$$v_p(\text{Disc } K) = \frac{|G|}{|I_0|} \sum_{i \geq 0} (|I_i| - 1). \quad (6.3)$$

The factor $|G|/|I_0|$ accounts for the regular representation, while the sum is the local different of the completed Galois extension. In particular, D_p does not occur.

TABLE 8. Regular-representation discriminant calculation by inertia type.

type	$(I_i)_i$	$\sum_i (I_i - 1)$	$51840/ I_0 $	$v_p(\text{Disc } K)$
$p = 3$	$(18, 9, 3, 3, 3, 3, 3, 1)$	37	2880	106560
$p = 5$	$(20, 5, 5, 5, 1)$	31	2592	80352
tame C_3	$(3, 1)$	2	17280	34560
reflection C_2	$(2, 1)$	1	25920	25920

For example, at 3 the sum is

$$(18 - 1) + (9 - 1) + 6(3 - 1) = 37,$$

and $2880 \cdot 37 = 106560$. At 5, $(20 - 1) + 3(5 - 1) = 31$, and $2592 \cdot 31 = 80352$.

Proposition 6.4 (Discriminant of the normal closure). *One has*

$$\text{Disc } K = 3^{106560} 5^{80352} A^{34560} B^{25920}.$$

This formula is independent of the alternatives $(D, I) = (140, 140)$ and $(206, 140)$ at 3.

Proof. Insert the four proved filtration types in [equation \(6.3\)](#); [table 8](#) lists every arithmetic step. The eight-prime support is exact by [proposition 3.1](#). Since only I_i appears in [equation \(6.3\)](#), the order of the decomposition overgroup at 3 is irrelevant. These calculations first give $|\text{Disc } K|$. Because $K/\mathbf{Q}$ is Galois of degree 51840, it is either totally real or totally imaginary; in the latter case $r_2(K) = 25920$ is even. Hence $\text{sgn}(\text{Disc } K) = (-1)^{r_2(K)} = 1$, proving the displayed positive formula. $\square$

The prime-power formulas, rather than expanded decimal integers, are the mathematical statements. For transport checking, the decimal value of $\text{Disc } E$ has 586 digits and that of $\text{Disc } K$ has 1,931,353 digits; their byte guards are recorded only in [appendix D](#).

7. INFINITY, EXACT SCOPE, AND VALIDATION BOUNDARY

7.1. Archimedean type. The maximal-order field computation gives

$$(r_1(E), r_2(E)) = (3, 12). \quad (7.1)$$

Thus complex conjugation fixes three of the 27 embeddings and exchanges the remaining 24 in pairs: its line type is $1^3 2^{12}$. Independently, exact Sturm counting gives four real roots of θ_{36} , hence double-six type $1^4 2^{16}$.

The simultaneous order-two table is [table 6](#). The line type alone would permit both ToM3 and ToM5; the double-six type selects ToM5. The finite-field class conventions in Banwait et al. [1, Table 7.1] provide useful context for $W(E_6)$ labels, but the match here comes from the two exact real-root counts and the reconstructed actions.

Proposition 7.1 (Complex conjugation). *The subgroup generated by complex conjugation is the Table-of-Marks subgroup ToM5. Under CTblLib 1.3.1, complex conjugation itself lies in element-class index 17 of CharacterTable("U4(2).2"); that class has size 540 and centralizer order 96. Its plus/minus dimensions are*

$$V_6 : (d^+, d^-) = (3, 3), \quad V_{20} : (d^+, d^-) = (11, 9).$$

Proof. Among the four order-two subgroup profiles, only ToM 5 has the pair of carrier types $1^3 2^{12}$ and $1^4 2^{16}$. Its fixed dimensions on (V_6, V_{20}) are $(3, 11)$. Since complex conjugation has order two, these are the plus dimensions; subtracting from 6 and 20 gives the minus dimensions. A separate reconstruction in the character table finds one order-two element class with size 540 and centralizer 96, namely index 17. $\square$

The first-use convention is therefore: *Table-of-Marks subgroup* ToM 5 / *character-table element-class index* 17 (*CTblLib* 1.3.1). The integers 5 and 17 belong to different indexing systems. Neither is a root number.

7.2. What the theorem does not determine.

`NO_BAD_EULER_OR_ROOT_NUMBER`. This paper proves no decomposition Frobenius, bad Euler polynomial or factor, local epsilon factor, local or global root number, Artin holomorphy, automorphy, analytic continuation, or functional equation. The alternatives $(D, I) = (140, 140)$ and $(206, 140)$ at 3 give identical filtered inertia, Artin–Swan conductors, representation conductors, and field discriminants. Even a future choice between them would not by itself authorize any of the excluded conclusions.

The firewall is a mathematical dependency statement. [Equations \(2.4\)](#) and [\(6.3\)](#) use only the filtration I_i , so the two possible D_3 overgroups are irrelevant to every quantity computed here. A bad Euler factor instead depends on a lift of residue Frobenius through D_3/I_0 , and epsilon factors require additional representation-theoretic and additive-character data. None of these inputs is reconstructed by the present calculations.

Two other boundaries prevent local evidence from being silently enlarged. First, θ_{36} is the only certified degree-36 local authority; δ_{36} is a bounded nonresult and contributes neither a premise nor corroboration. Second, the reflection primes have geometric ordinary-double-point/Picard–Lefschetz certification; no (e, f) decomposition row is claimed. The paper also proves no rational-point, local-point, weak-approximation, Hasse-principle, or Brauer–Manin assertion for Y , and no theorem for all cubic surfaces or all line fields.

7.3. Auxiliary exact validation. The finite calculations entering the propositions above admit an independent successful reconstruction from two immutable evidence carriers under a strict schema. The claim-to-artifact map, software versions, digest guards, and semantic perturbations are given in [appendix D](#). Those operational records validate the exact inputs; they are not a separate mathematical contribution and do not replace any written bridge in [sections 3](#) to [6](#).

8. CONCLUSION

For the explicit cubic surface [equation \(2.1\)](#), the ramification of the 27-line field is now determined at the level of lower inertia groups rather than only by its support or total discriminant. The key local fact is the 3-adic classification: a complete decomposition/inertia inventory, the ToM 6×2 , ToM 7, and ToM 8 branch-different exhaustion, and Serre’s odd-grade conjugation law leave one filtered inertia. The analogous normality and layer equations settle 5, while certified double-six partitions distinguish the tame C_3 class and exact ordinary-double-point geometry gives root reflections at the remaining primes.

These filtrations determine all local Swan and Artin conductors of (V_6, V_{20}) . Their products recover the degree-27 discriminant, and the regular representation gives the 1,931,353-digit normal-closure discriminant in a compact prime-power factorization. The same two permutation carriers distinguish complex conjugation and yield the archimedean plus/minus dimensions.

One local question remains deliberately open: the decomposition group at 3 has order 18 or 36. Both options contain the same filtered inertia ToM 140, so they change none of the results proved here. They also do not furnish bad Euler, epsilon, or root-number data.

Resolving that overgroup would require new local evidence, and any analytic consequence would require further independent input.

Two natural continuations are therefore separate from the present theorem. One is to recover the missing decomposition information at 3 without weakening the exact local-field standard used here. The other is to compute equally complete filtered-inertia packages for additional explicit full- $W(E_6)$ cubic surfaces, allowing genuine comparison of wild embeddings rather than comparison of bad-prime lists alone.

APPENDIX A. FROZEN INPUTS, LOCAL ORDERS, AND AUTHORITY BOUNDS

A.1. Exact algebraic input. The primitive coefficient array of [equation \(2.1\)](#) has 20 entries. The four Grassmann-chart equations are obtained by the direct substitution

$$(u_0, u_1, u_2, u_3) = (s, t, as + ct, bs + dt)$$

and equating the coefficients of s^3, s^2t, st^2, t^3 . A degree-order basis has Hilbert counts $(1, 4, 10, 12, 0)$; lexicographic conversion produces an eliminant $g(d)$ of degree 27 and rational back-substitutions for a, b, c . Substituting those expressions into all four chart equations, clearing denominators, and reducing modulo g gives four zero remainders. The complement of the chart is empty by five independent unit-ideal calculations.

The maximal-order calculation uses a transformed monic degree-27 polynomial together with an oriented isomorphism back to the original class of d . The orientation is part of the input: the original eliminant generator maps to a rational polynomial in the transformed generator. The full numerator vector is too large for print, so the following canonical guards identify it and the integral basis without abbreviation at the data layer.

TABLE 9. Compact guards for the exact field presentation.

object	SHA-256 or exact property
oriented original-generator image	aea67936d6dd88a7ef9ecaabd4740b737e52de9c34958e77bcefc5a51eed0e23
transformed monic polynomial	1ecdb17baaa586a5361cfd71ae6af218e05adb0b784aba514410a933fc74c64
canonical 27-element integral basis	5102fd87524498ef170b944f174b9c917279f7498f59b372de85b9913a0fbcc8
PARI textual basis carrier	ec66da41c6ad35e441c9916d3e448206ecc36e857fbab814475420a25c5d4813; 4,549,955 bytes
maximality certification	no unresolved prime returned by <code>nfcertify</code>

The semantics of `nfbasis`, `nfinit`, `idealprimedec`, `nfcertify`, and `polsturm` are those of the PARI/GP user's guide [10]. The software reference does not certify any displayed output; exact carriers and independent reconstruction do.

A.2. Divided discriminant engines. The Macaulay construction uses a 56×56 matrix and a 24×24 extraneous block. Fraction-free determinant evaluation and an independent exact polynomial engine give the same quotient. Its decimal-newline guard is

$$2be931285c05779d59f6d7f9f7006bd87a3c36e72c4d57dab7315413a3672ed9,$$

and its prime factorization is exactly [equation \(3.1\)](#). The field discriminant, computed from the integral basis rather than this resultant, is positive with 586 digits and guard

$$7548db5eb3f1c5549d80f6125521e9f3c7f965fb39b7198a8d28f93e8f78d6ca.$$

The different origin of these two integers is why the nine-prime surface envelope and eight-prime field support remain separate throughout the paper.

A.3. Complete local-row summary. For readability, repeated rows are compressed only by their multiplicity.

TABLE 10. All maximal-order local rows used in the theorem.

p	multiplicity	degree	ef	e	f	d	fd
3	1	3	3	3	1	3	3
	1	6	6	6	1	7	7
	2	9	9	9	1	18	18
5	2	1	1	1	1	0	0
	3	5	5	5	1	7	7
	1	10	10	1	1	15	15
181	1	3	3	3	1	2	2
	1	6	3	2	2	2	4
	1	18	3	6	2	2	12
997	1	3	3	3	1	2	2
	1	6	3	2	2	2	4
	1	18	3	6	2	2	12
2346241	1	3	3	3	1	2	2
	1	6	3	2	2	2	4
	1	18	3	6	2	2	12

The prime-ideal HNF carriers and prime-vector complements are retained in the arithmetic evidence rather than typeset: they occupy several megabytes and do not shorten any proof. Their canonical aggregate guard is

ee43a97eb36fdfb21565b339da78b7b2a4e73ef226e34b49223f127a1617ae4a.

A.4. All certified θ_{36} factor rows. At 3 and 5, the factor rows remain identical at precisions 900, 950, 1000, every factor is monic and simple, and the minimum coefficientwise multiply-back valuations are exactly 900, 950, 1000. The compressed rows are:

TABLE 11. Wild θ_{36} factor rows. The last column is the polynomial-discriminant valuation of one factor.

p	count	factor degree	mod- p exponent	residual degree	factor poldisc exponent
3	3	3	3	1	11
	1	9	9	1	62
	1	18	18	1	269
5	1	1	1	1	0
	1	5	5	1	27
	3	10	10	1	123

Thus the global/twice-largest bounds are 886/538 at 3 and 746/246 at 5. All three precisions exceed both numbers.

At each of 181, 997, 2346241, the θ_{36} rows are

factor degree	3	6	9	18
e	3	3	3	3
f	1	2	3	6
d	2	2	2	2
factor poldisc exponent	2	4	6	12

(A.1)

Their field-different contributions sum to 24, the global polynomial-discriminant exponent is 24, and twice the largest factor exponent is 24. Precision 40 clears both bounds, and factor multiplication is verified at 20, 30, 40.

For δ_{36} , the same apparent factor degrees at tame precision have factor polynomial-discriminant exponents 4, 20, 48, 204. Hence the twice-largest bound is 408, while the global exponent is 840. Precision 40 clears neither. No δ_{36} row, at any prime, is used to prove or corroborate a local partition. This negative authority statement is part of the premise graph, not an observation omitted from the successful lane.

APPENDIX B. EXHAUSTIVE SUBGROUP AND FIXED-SPACE LEDGERS

This appendix records the finite-group part of the proof independently of the local-arithmetic carriers in [appendix A](#). Throughout, “ToM” means the fixed ordering of the 350 conjugacy classes in `TableOfMarks("U4(2).2")`. A label is therefore a reproducibility coordinate, not an intrinsic name for a subgroup. Every retained class was reconstructed as a permutation subgroup on the labelled sets of 27 lines and 36 double-sixes. The GAP, TomLib, and CTblLib conventions are those recorded in [2, 6, 9]; none of those software references supplies an instance calculation for the surface.

B.1. Search predicate and branch contributions. For a subgroup $H \leq W(E_6)$, write

$$\lambda_{27}(H) = (|\Omega| : \Omega \in H \setminus \{1, \dots, 27\}), \quad \lambda_{36}(H) = (|\Xi| : \Xi \in H \setminus \{1, \dots, 36\}),$$

with each tuple sorted increasingly. The first pass compares these tuples with the exact local degrees. A proposed ordered pair (D, I) is then kept only when $I \trianglelefteq D$, the prescribed wild Sylow subgroup is normal in I , and D/I is cyclic. At $p = 3$ we additionally require I/P to be cyclic of order prime to 3, where P is the wild Sylow subgroup. These tests are run over all ordered containments between the 350 reconstructed classes; no abstract subgroup name is used as a search seed.

The branch-different calculation can be stated directly on the line action. Let $\Omega_1, \dots, \Omega_t$ be the I_0 -orbits. For $H \leq I_0$, put

$$c_j(H) = \frac{|H|}{|I_0|} (|\Omega_j| - \#(H \setminus \Omega_j)). \quad (\text{B.1})$$

Thus one lower layer equal to H contributes $(c_1(H), \dots, c_t(H))$ to the branch different vector. Formula [equation \(B.1\)](#) is simply the Artin conductor formula for the permutation module, separated over the I_0 -orbits. In particular its entries are rational before the complete filtration is assembled; they must not be rounded.

B.2. The complete $p = 3$ ledger. The simultaneous target is

$$\lambda_{27} = (3, 6, 9, 9), \quad \lambda_{36} = (3, 3, 3, 9, 18).$$

The raw and containment passes give exactly the following data.

There are no other raw hits: they are ToM 140, 142, and 206 only. The last class cannot be inertia because its quotient by its order-9 wild subgroup has order 4 and is not cyclic. It is retained in [table 12](#) solely as a decomposition overgroup.

For every occurrence of a normal order-3 subgroup compatible with one of these four pairs, [equation \(B.1\)](#) gives the following complete multiset. Keeping the multiplicity two for ToM 6 is necessary: the two entries come from different ordered containments.

TABLE 12. All $p = 3$ simultaneous hits and all valid ordered pairs.

D ToM	$ D $	$\text{IdGroup}(D)$	I ToM	$ D/I $	status before deep scan
140	18	(18, 4)	140	1	valid
142	18	(18, 3)	142	1	valid
206	36	(36, 10)	140	2	valid; D only
206	36	(36, 10)	142	2	valid; D only

TABLE 13. Full exact-rational deep-profile calculation at $p = 3$.

deep Q	multiplicity	line type	double-six type	$\dim(V_6^Q, V_{20}^Q)$	$\mathbf{c}(Q)$	$\mathbf{b}$	(r, s)
ToM 6	2	3^9	3^{12}	(0, 8)	$(\frac{1}{3}, \frac{2}{3}, 1, 1)$	(2, 5, 8, 8)	(7, -18)
ToM 7	1	$1^9 3^6$	$1^6 3^{10}$	(4, 10)	(0, 0, 1, 1)	(2, 5, 8, 8)	(1, 6)
ToM 8	1	3^9	$1^3 3^{11}$	(2, 6)	$(\frac{1}{3}, \frac{2}{3}, 1, 1)$	(2, 5, 8, 8)	(7, -18)

Here the target is $(3, 7, 18, 18)$ and one $P = C_3^2$ layer contributes $(1, 2, 4, 4)$. Solving

$$(2, 5, 8, 8) + r(1, 2, 4, 4) + \text{sc}(Q) = (3, 7, 18, 18)$$

over exact fractions produces the last column. Only ToM 7 gives a nonnegative integral solution. It gives one P layer and six Q layers. The grade-7 inversion test of [proposition 4.3](#) then removes the two rows with inertia ToM 142, leaving $(D, I) = (140, 140)$ and $(206, 140)$. This last test changes neither the deep profile nor the unresolved order of D .

B.3. The complete $p = 5$ and involution ledgers. At $p = 5$ the target pair is

$$(1, 1, 5, 5, 5, 10), \quad (1, 5, 10, 10, 10, 10).$$

All raw simultaneous hits and the normal-Sylow test are reproduced below.

TABLE 14. All $p = 5$ hits; the first row is the unique valid inertia class.

ToM	order	IdGroup	Sylow 5 normal?	conclusion
147	20	(20, 3)	yes	$(D, I, D/I) = (147, 147, 1)$
247	60	(60, 5)	no	excluded as inertia
295	120	(120, 34)	no	excluded as inertia

The base branch vector is $(0, 0, 4, 4, 4, 9)$, one C_5 layer contributes $(0, 0, 1, 1, 1, 2)$, and the target is $(0, 0, 7, 7, 7, 15)$. Hence precisely three positive layers occur. This recovers the lower orders $(20, 5, 5, 5, 1)$ without importing a break from the abstract group.

For completeness, the exhaustive order-two scan contains four classes. The two permutation profiles, rather than order alone, distinguish a root reflection from complex conjugation.

TABLE 15. All order-two subgroup classes in the two labelled actions.

subgroup	ToM	line type	double-six type	$\dim(V_6^H, V_{20}^H)$	$ N_G(H) $	class size	centralizer	element index
2		$1^{15} 2^6$	$1^{16} 2^{10}$	(5, 15)	1440	36	1440	16
3		$1^3 2^{12}$	$1^{12} 2^{12}$	(2, 12)	1152	45	1152	2
4		$1^7 2^{10}$	$1^8 2^{14}$	(4, 12)	192	270	192	3
5		$1^3 2^{12}$	$1^4 2^{16}$	(3, 11)	96	540	96	17

For a subgroup of order two, its normalizer equals its centralizer. The reflection geometry selects subgroup ToM 2. At infinity the simultaneous line/double-six profile selects subgroup ToM 5; only after this subgroup selection does the character table identify element-class index 17.

B.4. Fixed spaces and conductor arithmetic. The fixed spaces were computed from the labelled rational matrices for V_6 and V_{20} , not inferred from subgroup order. The complete list used in the paper is

place or role	H	$\dim(V_6^H, V_{20}^H)$	$\text{codim}(V_6^H, V_{20}^H)$
$p = 3, I_0$	ToM 140	$(0, 3)$	$(6, 17)$
$p = 3, I_1$	C_3^2	$(0, 4)$	$(6, 16)$
$p = 3, I_2 = \cdots = I_7$	ToM 7	$(4, 10)$	$(2, 10)$
$p = 5, I_0$	ToM 147	$(2, 3)$	$(4, 17)$
$p = 5, I_1 = I_2 = I_3$	C_5	$(2, 4)$	$(4, 16)$
tame C_3	ToM 6	$(0, 8)$	$(6, 12)$
reflection	ToM 2	$(5, 15)$	$(1, 5)$
complex conjugation	ToM 5	$(3, 11)$	$(3, 9)$

(B.2)

Substitution into the lower-numbering formula gives, with no omitted layer,

$$\begin{aligned} \text{Sw}_3 &= \frac{9}{18}(6, 16) + 6\frac{3}{18}(2, 10) = (5, 18), \\ \text{Sw}_5 &= 3\frac{5}{20}(4, 16) = (3, 12). \end{aligned}$$

Adding the I_0 codimensions gives $(11, 35)$ and $(7, 29)$, respectively. For the tame rows the Swan term is zero and the final column of [equation \(B.2\)](#) is already the Artin pair.

Finally, the normal-closure exponents provide a group-order check that is manifestly independent of D_3 . For a Galois extension with group G and lower inertia groups I_i ,

$$v_p(\text{Disc } K) = \frac{|G|}{|I_0|} \sum_{i \geq 0} (|I_i| - 1). \quad (\text{B.3})$$

With $|G| = 51840$, the four local sums and multipliers are

p	$\sum_i (I_i - 1)$	$ G / I_0 $	$v_p(\text{Disc } K)$
3	37	2880	106560
5	31	2592	80352
181, 997, 2346241	2	17280	34560
283, 1801, q	1	25920	25920.

This calculation uses I_0 and its filtration only; the two possible decomposition-group orders at 3 therefore give the same result.

APPENDIX C. EXACT REFLECTION GEOMETRY AT THE THREE LARGE-SUPPORT PRIMES

We give the instance part of the ordinary-double-point argument used in [section 5.4](#). This appendix deliberately contains no local factorization or (e, f) row: the output is an inertia class obtained from geometry.

C.1. Singular loci on four disjoint pivot strata. For $j = 0, 1, 2, 3$, let the j th pivot stratum be

$$u_0 = \cdots = u_{j-1} = 0, \quad u_j = 1,$$

with $u_{j+1}, \dots, u_3$ as its remaining coordinates. These strata are disjoint and exhaust projective space. Over $\mathbf{F}_p$ we specialize the four homogeneous first derivatives of F to each stratum and compute their reduced Gröbner basis. Since $p \neq 3$, Euler's identity

implies that a common gradient zero also lies on $F = 0$. In every case below, the stratum-0 ideal has the displayed reduced linear basis, whereas the ideals on strata 1, 2, 3 reduce to the unit ideal. Thus the projective singular locus consists of exactly one reduced point.

TABLE 16. Exact pivot-stratum-0 singular-locus carriers.

p	reduced stratum-0 basis	unique projective point P_p
283	$x_1 + 217, x_2 + 128, x_3 + 158$	$[1, 66, 155, 125]$
1801	$x_1 + 364, x_2 + 1263, x_3 + 1290$	$[1, 1437, 538, 511]$
q	$x_1 + 1421943369344251325350771506077$ $769703165460134637, x_2 + 512738467958$ $6422310072147954359368763636216685$ $431, x_3 + 128720431192486112814927185$ 03093548885792203018845	$[1, 135101038131290406705$ 09336985505882430772547 $129082, 9804662502886869$ 68578796053722428337030 $1790578288, 206000406322$ 46807143673899884901032 $48145804244874]$

The coordinates in the third row are reduced to the interval $[0, q - 1]$; the linear-basis constants are their negatives modulo q . The unit-ideal outcomes on the other three pivot strata complete the projective exhaustion; they are not overlapping affine-chart tests of the same point.

C.2. Hessians and first Hensel corrections. Put $f = f_0 = F(1, x_1, x_2, x_3)$. The affine Hessian matrices at the points of table 16, reduced modulo p , are

$$H_{283} = \begin{pmatrix} 146 & 136 & 91 \\ 136 & 13 & 203 \\ 91 & 203 & 263 \end{pmatrix},$$

$$H_{1801} = \begin{pmatrix} 276 & 326 & 379 \\ 326 & 700 & 902 \\ 379 & 902 & 1796 \end{pmatrix},$$

Write $H_q = (h_{ij})_{1 \leq i, j \leq 3}$, with $h_{ij} = h_{ji}$. Its six independent entries are

$$\begin{aligned} h_{11} &= 1311544089871005361766843220403139014755424019851, \\ h_{12} &= 9978936465940142019654684549662337052767937590180, \\ h_{13} &= 8612858348525205055038179080298823074941248581837, \\ h_{22} &= 2137397871150289173057311598746005216526412381550, \\ h_{23} &= 253346114695653176136051595749839977708789738244, \\ h_{33} &= 14594568215694396621068754907779675489233126539611. \end{aligned}$$

Their determinants modulo p are respectively

$$228, \quad 1387, \quad 6136116089260018682592250996037036352166217747437, \quad (\text{C.1})$$

and are nonzero. Consequently each singularity is an ordinary quadratic singularity, and the gradient map has a unique zero lifting P_p modulo p^2 .

For transparency, write

$$\mathbf{g}_p = \frac{\nabla f(P_p)}{p} \pmod{p}, \quad H_p \mathbf{c}_p = -\mathbf{g}_p \pmod{p}.$$

The exact reduced vectors are

$$\begin{aligned} \mathbf{g}_{283} &= (141, 174, 163), & \mathbf{c}_{283} &= (222, 71, 239), \\ \mathbf{g}_{1801} &= (508, 508, 1167), & \mathbf{c}_{1801} &= (495, 192, 1792). \end{aligned}$$

For the large prime, the two triples are

$$\begin{aligned} \mathbf{g}_q &= (3555487038157581200031692877049198796770382105104, \\ &\quad 8802808374986467697053812449802016955221345926655, \\ &\quad 14062402405382925912884193140669738406166578327338), \\ \mathbf{c}_q &= (2425745955169956924323135015855316312606784196721, \\ &\quad 10534474137669716049158008564147639826733568383236, \\ &\quad 14797023225454273436443782496570083367866547475252). \end{aligned}$$

The critical lift is the exact residue class

$$\tilde{P}_p = [1, \mathbf{x}(P_p) + p\mathbf{c}_p] \pmod{p^2}. \quad (\text{C.2})$$

For the two smaller primes this gives

$$\tilde{P}_{283} = [1, 62892, 20248, 67762], \quad \tilde{P}_{1801} = [1, 892932, 346330, 3227903].$$

Equation (C.2), together with the fully printed P_q and $\mathbf{c}_q$, is an exact (and more legible) carrier for the q^2 lift than its three expanded 98-digit coordinates.

C.3. Critical value and transversality. Direct integer evaluation gives the following unit residues:

p	$F(P_p)/p \pmod{p}$	$\det H_p \pmod{p}$
283	212	228
1801	818	1387

(C.3)

$$F(P_q)/q \equiv 11651769163508833344099877335703302197941640200357 \pmod{q},$$

$$\det H_q \equiv 6136116089260018682592250996037036352166217747437 \pmod{q}.$$

Because $\nabla f(P_p)$ is divisible by p , Taylor expansion and equation (C.2) give

$$f(\tilde{P}_p) \equiv f(P_p) \pmod{p^2}.$$

The critical-value column of equation (C.3) is nonzero, hence $v_p(f(\tilde{P}_p)) = 1$. After the unique critical-point translation, the completed local equation has a nondegenerate quadratic term and a smoothing parameter equal to p times a unit. It follows that the total space is regular at the critical point and that the base trait meets the discriminant divisor transversely.

All three residue characteristics are odd. The Picard–Lefschetz formula of SGA 7 II, Exposé XV [3], together with the divided-discriminant bridge for cubic surfaces in Saito [7, Proposition 2.3, p. 858; Theorem 3.5, pp. 864–866; cubic-surface specialization, p. 870], therefore makes tame inertia act by a single root reflection. The exhaustive action comparison in table 15 identifies its subgroup as ToM 2, with line type $1^{15}2^6$, double-six type $1^{16}2^{10}$, fixed dimensions (5, 15), and Artin pair (1, 5). This conclusion uses no residue-degree assertion.

APPENDIX D. EXACT CERTIFICATE AND ARTIFACT MAP

This is the only artifact map in the paper. Its purpose is to make the large exact carriers addressable without confusing a machine record with a mathematical authority. The universal implications used in the proofs are cited from the literature; the records below bind the arithmetic and group computations for this one surface.

D.1. Claim-to-carrier map. The certificate payload is divided into eight semantic gates, G0–G7. For the reader, the following coarser map C1–C7 groups each mathematical claim with the carrier that permits an independent reconstruction.

TABLE 17. The complete C58 artifact map.

map	mathematical content	independent reconstruction carrier
C1	frozen cubic, Grassmann elimination, smoothness, degree-27 field, and oriented field presentation	canonical surface and exact-field leaves in <code>c58_arithmetic_evidence.json.gz</code> ; independent surface and field reconstruction in the checker
C2	divided discriminant, maximal order, local prime ideals, differentials, $\text{Disc}(E)$, and the nine-versus-eight support distinction	arithmetic evidence, strict local-row schema, prime-vector complements, and independent PARI reconstruction
C3	the certified θ_{36} partitions at precisions 20, 30, 40 and 900, 950, 1000, including separation bounds and multiply-back; δ_{36} nondependency	arithmetic evidence plus the named authority and nonresult leaves in the certificate; both lanes are rebound independently
C4	the 350-class subgroup scan, every $p = 3$ and $p = 5$ raw hit and valid (D, I) pair, the ToM $6 \times 2, 7, 8$ rational ledger, normality, and the grade-7 action	<code>c58_group_evidence.json</code> , reconstructed labelled actions, exact rational arithmetic, and a separate GAP checker
C5	four-stratum reflection geometry, Hessians, Hensel corrections, critical values, transversality, and the ToM-2 selection	arithmetic reflection witnesses together with the complete order-two profile map in the group evidence
C6	fixed spaces, all Swan and Artin conductors, $N(V_6)$, $N(V_{20})$, $\text{Disc}(E)$, $\text{Disc}(K)$, signature, subgroup ToM 5, and element index 17	independent character, product, signature, and CTblLib reconstruction leaves
C7	strict parsing, schema closure, source/result inventory, checker independence, hostile scalar rebound, and scope firewalls	certificate, schema, independent check report, self-excluding scoped manifest, and the project-local test report

The complete payload has 1149 scalar leaves. The checker reconstructs all semantic leaves without importing or calling the producer’s theorem helpers, and rejects 1199 systematic rebound mutations. Gates G0–G7 all pass; the project-local scaffold suite contains 45 passing tests. These are integrity statements about the frozen computation, not statistical evidence.

D.2. Exact identities. The principal SHA-256 identities are

TABLE 18. Principal certificate identities.

object	SHA-256
results/c58_certificate.json	456a481368d593f0d015436bf8a3a518d15b4567880fa7726c77d29a259d79ee
canonical payload	fba2dfdf71977d8de6c85635eca6572e0b8a0680570f394af9e3e9e8698f732f
results/c58_schema.json	ccbc20eb6e04d00f14cdc0ccf970caebf4d66b4103176515799ddca89639009a
results/c58_check_report.json	64454700ddaa0bb9ff56c85afa213f038ec6b430bc38ef07e3f22924081d22e9
independent replay summary	d5e98eaafedd03b90c851f9082237387f1eddc559a8c37b49b85f0e5ffa897c6
results/c58_arithmetic_evidence.json.gz	e374d328a7937c48af93e0b46f54eead5a878f01acc161d8053fe4a10c5f6128
decompressed canonical arithmetic JSON	816b8f11358c266826c3ab7117705e5d8fc2426ad373c08a47026d0ee534cd3d
results/c58_group_evidence.json	0e0b3fd4927b3a8355037b57b86a1e3cc7efe15832be4f5ca76cb4989b71a1fd
results/scoped_hash_manifest.json	a18742298722e2bffa022b95be8a09806dd774a52ab8e095ebde78924c45ae730

The scoped tree contains exactly the following 14 source files:

README.md, c58_arithmetic.py, c58_atomic_promote.py, c58_checker.py, c58_checker_group.g, c58_checker_pari.py, c58_exact.py, c58_group.py, c58_hash_manifest.py, c58_pipeline.py, c58_producer.py, c58_surface.py, run_all.sh, and test_c58.py.

Its eight result files are

RESULTS.md, TEST_REPORT.md, c58_arithmetic_evidence.json.gz, c58_certificate.json, c58_check_report.json, c58_group_evidence.json, c58_schema.json, and scoped_hash_manifest.json.

Thus the live code/results inventory has 22 entries. The manifest is self-excluding and contains the other 21 entries, so its own digest is not circularly embedded in its contents.

D.3. Backend and replay contract. The bound backend versions are

backend	versions
PARI lane	Python 3.10.12, cypari2 2.1.2, PARI 2.13.3
finite-group lane	GAP 4.11.1, TomLib 1.2.9, SmallGrp 1.4.1, CTblLib 1.3.1
exact-polynomial lane	Python 3.12.3, FLINT 0.9.0, SymPy 1.14.0, jsonschema 4.25.0.

In particular, a CTblLib version change invalidates the numeric character-table labels until a fresh rebinding is performed.

From the project-local code directory, the unique nonmutating command is

```
/usr/bin/bash -p ./run_all.sh
```

invoked by a trusted parent after unsetting loader, shell-startup, Python optimization, and Python import-path variables. The trust contract excludes a concurrent same-UID pathname mutator during the external-child launch window. A deliberate refresh is a different maintainer operation and is not needed to check the theorem. It promotes exactly

six generated targets as a rollback-safe group and then requires a second clean default replay.

The canonical JSON parser rejects duplicate keys and noncanonical integer encodings; deterministic gzip binds the compressed and decompressed arithmetic records. The independent checker redoes local decompositions, different exponents, subgroup filters, filtrations, conductors, and global identities. Raw exploratory transcripts and undocumented temporary outputs are outside the premise graph.

Finally, the payload includes the exact scope leaf NO-BAD-EULER-OR-ROOT-NUMBER. It rejects any output asserting a decomposition Frobenius, bad Euler polynomial or factor, epsilon factor, local or global root number, Artin holomorphy, automorphy, analytic continuation, or functional equation. The unresolved order of D_3 does not weaken any result in [theorem 2.1](#), and resolving it would not by itself authorize any of those excluded statements.

APPENDIX E. SOURCE-USE LEDGER AND THEOREM BOUNDARIES

The proof separates general theorems from computations for the frozen surface. The exact source locator, the implication used, and the assertion that remains instance-specific are recorded in [table 19](#). This prevents a citation to a general theorem from being read as a citation for one of the large exact outputs.

TABLE 19. Complete source-use ledger.

source and locator	authorized use here	not supplied by the source
Serre, <i>Local Fields</i> , Chapter II, §2, Exercises 1–2, printed p. 30 [8]	Krasner–Hensel factor stability once the separation and precision inequalities have been proved	the local factors, polynomial-discriminant exponents, precision bounds, or multiply-back identities for θ_{36}
Serre, Chapter IV, §2, Proposition 9, printed pp. 69–70 [8]	the graded tame action $\theta_i(s\tau s^{-1}) = \theta_0(s)^i \theta_i(\tau)$	the 350-class search, the ToM $6 \times 2, 7, 8$ profile inventory, or the identification of ToM 140 before the grade-7 test
Serre, Chapter VI, §2, Corollary 1', printed pp. 100–101; Chapter VI, §3, Proposition 6 and Corollary 1, printed pp. 103–104 [8]	Artin conductors; conductor–discriminant identity	any local fixed space, Swan value, conductor exponent, or field discriminant for this instance
SGA 7 II, Exposé XV [3]	Picard–Lefschetz reflection monodromy for a transverse ordinary quadratic singularity	the singular points, Hessians, Hensel lifts, transversality checks, or the ToM label at 283, 1801, q

source and locator	authorized use here	not supplied by the source
Saito, Proposition 2.3, p. 858; Theorem 3.5, pp. 864–866; cubic-surface specialization, p. 870 [7]	the divided discriminant, determinant character, and Picard–Lefschetz bridge for cubic surfaces	the frozen divided discriminant, the three local witnesses, higher ramification groups, or either field discriminant
Elsenhans–Jahnel, Propositions 21–22, pp. 651–652 [4]	precedent for explicit full- $W(E_6)$ cubic surfaces, ramification envelopes, and p -adic factor patterns	the filtered wild inertia, Swan conductors, maximal-order discriminant, or normal-closure discriminant of the present surface
Elsenhans–Jahnel, Theorem 2.12 [5]	the cubic discriminant and its index-two determinant character	the complete local representation or any frozen local calculation
Banwait–Fité–Loughran, Table 7.1 [1]	the corrected finite-field $W(E_6)$ class/trace convention	number-field inertia, filtration, or conductor data for the present surface
PARI/GP user’s guide sections for <code>nfbasis</code> , <code>nfinit</code> , <code>idealprimedec</code> , <code>nfcertify</code> , and <code>polsturm</code> [10]	semantics of maximal orders, prime decompositions, differentials, certification, and real-root counting	proof that any particular displayed output was generated or independently reconstructed correctly
GAP/TomLib and CTblLib package records [2, 6, 9]	the reproducible U4(2).2 Table-of-Marks and character-table naming convention, including package versions	the frozen-instance subgroup classification, or permission to identify subgroup ToM 5 with element-class index 17

The closest published precedent therefore prevents a novelty claim of the form “the first ramification calculation for an explicit cubic surface.” The bounded contribution proved here is the complete filtered-inertia and Artin-conductor package for this explicit full- $W(E_6)$ surface, including both wild primes and the exact line-field and normal-closure discriminants. It is not a claim about all cubic surfaces.

Nor does the paper address rational or local points, weak approximation, the Hasse principle, or a Brauer–Manin obstruction for Y . Those questions are logically separate from the permutation ramification theorem. The analytic nonclaims are the stronger named firewall stated in [remark 2.3](#) and repeated in [appendix D](#).

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